RFI: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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1. Introduction

On behalf of the NSF Institute for Data-Driven Dynamical Design (ID4), we strongly support OSTP's effort to identify the key policy areas needed to sustain and enhance America's AI dominance. As leading researchers in the application of AI to challenges in the applied sciences and engineering, it is clear to us that AI is rapidly transforming the process of design and discovery and revealing foundationally new mechanisms. In this submission, we focus on two critical areas where we have significant expertise: (1) research and development in the context of applied science and engineering and (2) education and workforce development at the intersection of academia and industry.

2. About ID4

The Institute for Data-Driven Dynamical Design, funded under NSF's Harnessing the Data Revolution (HDR) initiative, exemplifies a transformative approach to research by deeply integrating applied science and computer science around AI solutions. The Institute comprises 16 research groups at 11 R1 universities and corporate partner. Further, we interact regularly with another dozen research groups in the US and abroad. ID4 aims to transform the design cycle for scientists and engineers. From chemistry to civil engineering, we seek to create platforms that accelerate the discovery of materials with new mechanisms and dynamics through the complementary union of human and machine intelligence. The 85 papers and codebases ID4 has developed in its first four years are available at: https://id4.mines.edu/papers-codes/. These developments focus on new AI techniques that respond to the unique needs of the applied sciences and engineering community. These developments range from autodifferentiable methods for engineering to corpus-informed decision making in the automated chemistry laboratories of our industrial partner. This integration enables breakthroughs otherwise unattainable within isolated domains. In parallel with our focus on methods development, we aspire to create the next generation of leaders in this transformative approach and to cultivate a community committed to data-driven approaches. Beyond training our own students, ID4 has run a multitude of NSF-sponsored workshops on AI-ready data, AI-powered data visualization, uncertainty quantification in machine learning, AI for materials science, and the annual workshop for the NSF Harnessing the Data Revolution ecosystem.

3. Research and Development Recommendations

Conduct a Holistic Review: Bring together stakeholders across current federal programs and institutes (e.g., HDR, AI Institutes, ERCs, MRSECs, DMREFs) to assess their current AI infrastructure and needs. Use findings to refine funding priorities and organizational models. For example, within the NSF HDR and NSF DMREF ecosystems, there are annual meetings; however, these meetings do not span between ecosystems and do not actively develop priority areas. This is in contrast to, for example, the nuclear physics community that makes a serious effort to define their strategic goals.

Support Specialized Research: Fund initiatives that develop physics-informed AI and interpretable models tailored for scientific and engineering applications. Examples include (i) research on axiomatic AI methods that integrate physical constraints directly into hypothesis generation and decision making, (ii) navigating massive design spaces with autodifferentiation, (iii) integrating human and machine insight into multi-objective decision making. In our Institute, we have found that the AI/ML that industry has focused on over the last two decades (e.g., computer vision, natural language processing) do not necessarily map well to the challenges of science and engineering. Instead, the major developments have been driven by specific scientific problems that generate new classes of solutions. One such example is the emergence of equivariant neural nets for learning how atoms interact, which is important across materials science, chemical engineering, and chemistry.

Develop Comprehensive Testing Environments and Model Registry: Create national AI testbeds that replicate real-world physical and industrial settings. These platforms would be used to evaluate model performance, robustness, and—crucially—uncertainty quantification, ensuring models are safe and resource-efficient. Part and parcel with these efforts is the development of benchmarks for AI in the applied sciences/engineering and the implementation of a central registry. This transparency would facilitate benchmarking against challenge datasets and provide a common language for evaluating model strengths and weaknesses. Within our Institute, we have developed a benchmark for anomaly detection in atomistic simulations motivated by the massive amount of data such simulations produce and the challenges of mapping complex objects across space and time. Likewise, we have developed benchmarks for LLMs for material property prediction (LLM4Mat-Bench).

Establish Dedicated Consortia: Create federally funded consortia that embed computer scientists directly into applied research projects alongside physical scientists, chemists, and engineers. This model—exemplified by ID4—ensures that AI innovations address real-world challenges from the outset. For example, within ID4 we have developed new approaches to lower building construction costs through the integration of auto-diffferentiation and symmetry-conserving continuous flows in the design of structural elements.

Pilot Automated Lab Initiatives: Encourage pilot projects where AI-driven hypothesis making and decision support systems are integrated with automated laboratory processes, thereby accelerating scientific discovery with built-in physical constraints. Such pilot projects could target industries that are lagging in the adoption of modern (post-2020) AI methods. Within ID4, we have a strong partnership with US company Kebotix. Kebotix uses automated laboratory facilities and AI to accelerate the discovery of new chemicals and synthetic pathways therein. This relationship has been bidirectional with academia: Kebotix has guided ID4 on emerging industry-specific problems and the academic side of ID4 has been sharing their emerging software involving decision making (e.g., Bayesian methods) and visualization. By beta-testing these software packages, Kebotix gains access to state-of-the-art software and provides feedback to the developers.

4. Education and Workforce Development Recommendations

Facilitate Regular Interdisciplinary Interactions: Support targeted workshops, fellowships, and collaborative funding mechanisms centered around grand challenges (e.g., materials in extreme environments or advanced manufacturing). These forums should include the development and use of challenge/benchmark datasets that provide standardized evaluation metrics for new AI methods. For example, ID4 hosted a national workshop on AI for data visualization techniques that had strong representation from both academia and the biomedical industry; the outcome of this workshop included grand challenges to help academia better serve industrial needs.

Promote Industry–Academic Partnerships: In the context of workforce development, AI-based industry-academic partnerships are vital for both providing continuing education to existing workers and for building a pipeline for students from academia to jobs that will value their AI expertise. Programs that fund industry-academic partnerships beyond the usual SBIR and GOALI models are vital. Within ID4, we enhance our collaboration with Kebotix by having undergraduate students from our departments conduct summer internships at Kebotix.

Integrate Interdisciplinary Curriculum: The continued investment in educational programs that combine computer science with physical sciences and engineering remains vital at all levels—from K-12 to graduate studies. To remain globally competitive in AI, we need to teach the next generation foundational skills ranging from data fluency to statistics to the practical integration of AI in scientific research. For example, the NSF REU program that adjoins ID4 (Integrating Computation and Experiment to Create Revolutionary Materials) has found that STEM majors in their senior year have typically had no exposure to AI/ML in their curriculum but at the same time are excited to learn. NSF REU and other similar programs have a proven track record of supporting development of a strong STEM workforce in the United States, with the majority of students who participate in these programs going on to pursue careers in STEM.

Support National Fellowships: Fund regular national AI-focused fellowship programs that promote continuous engagement between academia, industry, and government. These efforts will help nurture a technologically adept workforce capable of leveraging AI for transformative innovations. While the DOE Computational Science Fellowship is broadly a step in the right direction, all Federal agencies would benefit from graduate fellowships that focused student research on AI towards the agencies' directives.

5. Conclusion

The experiences within ID4 clearly illustrate the profound advantages of integrating applied sciences with computer science and industrial partners to help assure America's continued AI dominance. Continuing federal investment in these integrative research environments will undoubtedly produce revolutionary advancements across science, technology, and society, positioning future researchers to tackle increasingly complex and multidisciplinary challenges facing the United States.